Pre-Calculus II: Graded Worksheet: #6

Due on May 14, 2022 at 6:45pm

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Problem 1

Use the triangle given in Figure 1 to find $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$. Provide *exact* values. (The triangle may not be drawn to scale.) [5 **points**]

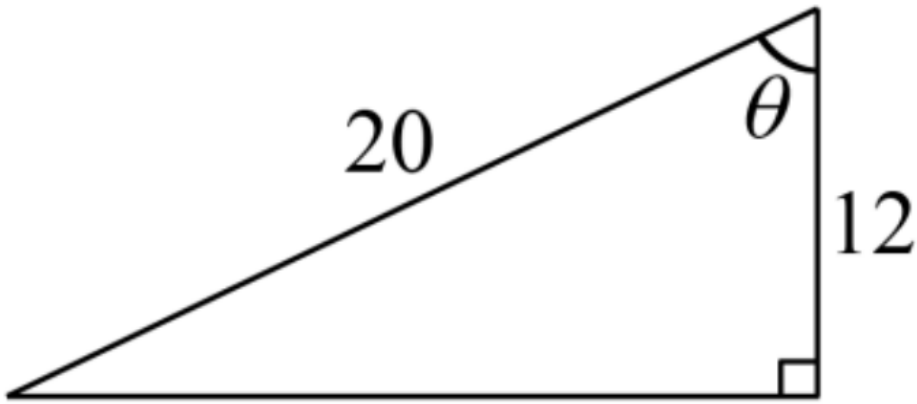

Figure 1

Solution 1

First, we're going to use the Pythagorean Theorem to find all the missing sides.

$$\begin{aligned} & a^2 + b^2 = c^2 \\ \Rightarrow \quad & a^2 + 12^2 = 20^2 \\ \Rightarrow \quad & a^2 + 144 = 600 \\ \Rightarrow \quad & a^2 = 456 \\ \Rightarrow \quad & \sqrt{a^2} = \sqrt{456} \\ \Rightarrow \quad & a = 2\sqrt{114} \end{aligned}$$

.

Now, we can solve for the 3 trig functions.

$$\begin{aligned} & \sin(\theta) = \frac{\text{OPP}}{\text{HYP}} \\ \Rightarrow \quad & \sin(\theta) = \frac{2\sqrt{114}}{20} \end{aligned}$$

.

So, $\sin(\theta) = \frac{2\sqrt{114}}{20}$

$$\begin{aligned} & \cos(\theta) = \frac{\text{ADJ}}{\text{HYP}} \\ \Rightarrow \quad & \cos(\theta) = \frac{12}{20} \\ \Rightarrow \quad & \cos(\theta) = \frac{3}{5} \end{aligned}$$

.

Problem 1 continued on next page. . .

So, $\cos(\theta) = \frac{3}{5}$

$$\begin{aligned} & \tan(\theta) = \frac{1}{\frac{\text{OPP}}{ADJ}} = \frac{\text{ADJ}}{\text{OPP}} \\ \Rightarrow \quad & \tan(\theta) = \frac{12}{2\sqrt{114}} \\ \Rightarrow \quad & \tan(\theta) = \frac{6}{\sqrt{114}} \end{aligned}$$

.

So, $\tan(\theta) = \frac{6}{\sqrt{114}}$

Problem 2

Solve the triangle by finding the missing side a and missing angles X and Y. (The triangle may not be drawn to scale.) Approximations are acceptable here, although they must be denoted correctly. [5 **points**]

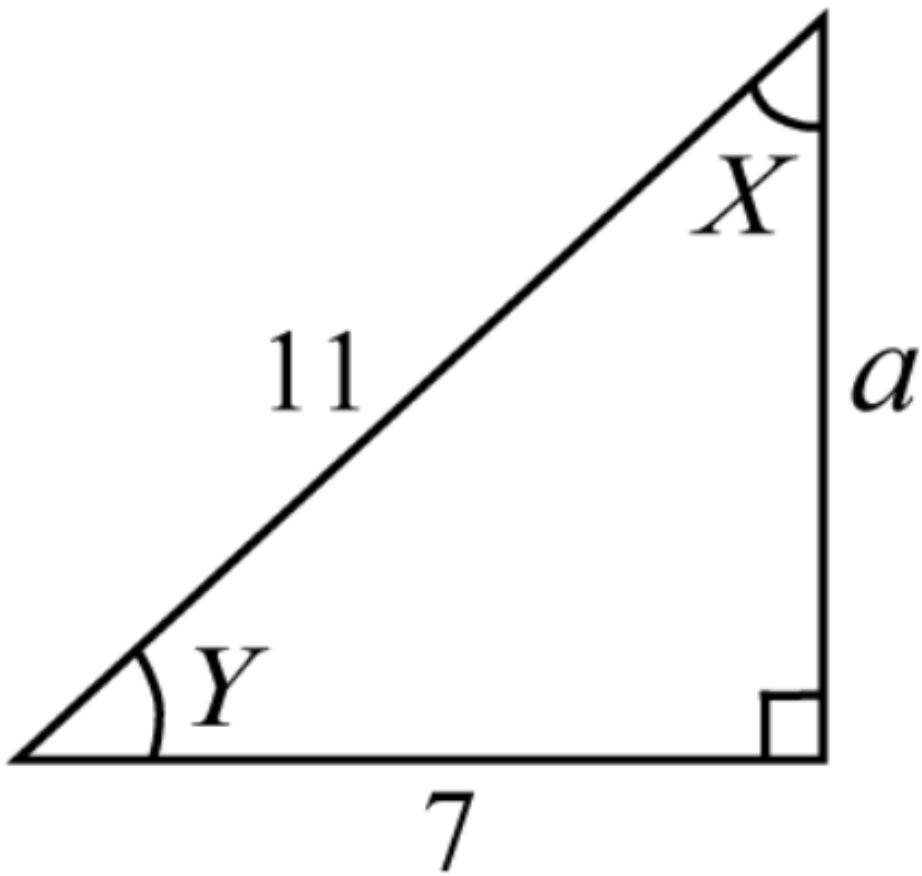

Figure 2

Solution 2

To solve for a, we can use the Pythagorean Theorem:

$$\begin{aligned} & a^2 + b^2 = c^2 \\ \Rightarrow \quad & a^2 + 7^2 = 11^2 \\ \Rightarrow \quad & a^2 + 49 = 121 \\ \Rightarrow \quad & a^2 = 72 \\ \Rightarrow \quad & \sqrt{a^2} = \sqrt{72} \\ \Rightarrow \quad & a = 6\sqrt{2} \end{aligned}$$

.

So, $a = 6\sqrt{2}$.

Problem 2 continued on next page. . .

To find our missing angles, we need to use the inverse functions. First, let's solve for X:

$$\begin{aligned} & \tan(Y) = \frac{6\sqrt{2}}{7} \\ \Rightarrow \quad & \tan^{-1}(\tan(Y)) = \tan^{-1}\left(\frac{6\sqrt{2}}{7}\right) \\ \Rightarrow \quad & Y \approx 0.88 \\ & . \end{aligned}$$

Now, we can use the 180° rule to find for X:

$$\begin{aligned} & 90 + 0.88 = 90.88 \\ \Rightarrow \quad & 180 - 90.88 = x \\ \Rightarrow \quad & X \approx 89.12^\circ. \end{aligned}$$

So, our final answers are:

$$\begin{aligned} a &= 6\sqrt{2} \\ X &\approx 89.12^\circ \\ Y &\approx 0.88^\circ \\ & . \end{aligned}$$

Problem 3

Solve the triangle by finding the missing side c and missing angles A and B. (The triangle may not be drawn to scale.) Approximations are acceptable here, although they must be denoted correctly. [5 **points**]

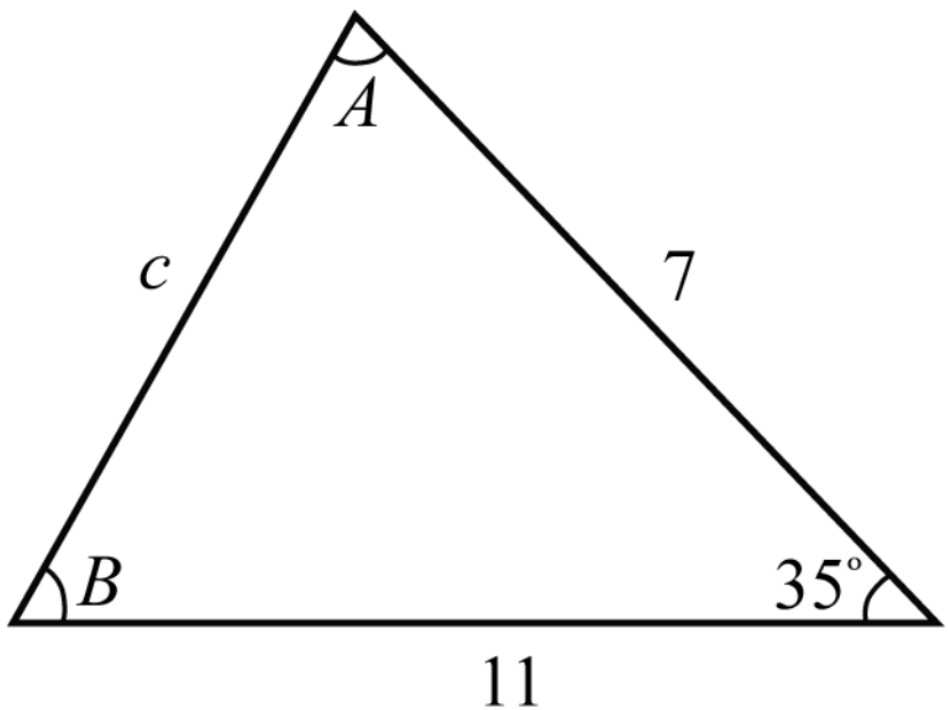

Figure 3

Solution 3

First, let me solve for c.

$$\begin{aligned} & c^2 = 11^2 + 7^2 - 2(11)(7) \times \cos(35^\circ) \\ \Rightarrow \quad & c^2 \approx 43.85 \\ \Rightarrow \quad & \sqrt{c^2} \approx \sqrt{43.85} \\ \Rightarrow \quad & c \approx 6.62 \end{aligned}$$

.

Now we can use the **Law of Sines** to find either of the other angles. Let's find B:

$$\begin{aligned} & \frac{\sin(B)}{7} = \frac{\sin(35^\circ)}{c} \\ \Rightarrow \quad & \sin(B) = \frac{7 \times \sin(35^\circ)}{c} \\ \Rightarrow \quad & \sin^{-1}(\sin(B)) = \sin^{-1}\left(\frac{7 \times \sin(35^\circ)}{c}\right) \\ \Rightarrow \quad & B \approx 37.34 \end{aligned}$$

.

Problem 3 continued on next page. . .

Now we can solve for A:

$$\begin{aligned} & A + B + 35^\circ = 180^\circ \\ \Rightarrow \quad & A + 37.34^\circ + 35^\circ = 180^\circ \\ \Rightarrow \quad & A + 72.34^\circ = 180^\circ \\ \Rightarrow \quad & A \approx 107.66^\circ \end{aligned}$$

.

So, here are our final answers:

$$\begin{aligned} A &\approx 107.66^\circ \\ B &\approx 37.34^\circ \\ c &\approx 6.62. \end{aligned}$$